Unit 13, Lesson 9: Identifying and Interpreting Parts of an Expression or Equation – Part 2

Benchmark

MA.912.AR.1.1 Identify and interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context, including viewing one or more of its parts as a single entity.

Learning Targets

- ☐ I can identify parts of an expression including factors, terms, constants, coefficients, and variables.
- ☐ I can interpret parts of an equation or expression that represent a quantity in terms of a mathematical or real-world context.

13.9.1 Warm-Up: Bell Work

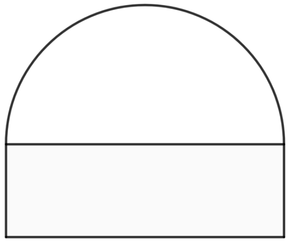
1. The Bureau of Economic Analysis tracks information on consumer spending. A data scientist working at the Bureau found that the function $P(t) = 5t + 84$ models the amount of spending, P, in billions of dollars, for t years since 2012.

   Complete the table.

Quantity	Description
5	○ initial value ○ rate of change ○ informational only
84	○ initial value ○ rate of change ○ informational only
2012	○ initial value ○ rate of change ○ informational only

13.9.2 Guided Instruction: Parts of an Equation or Expression

1. An architect is designing a window for a building. The window consists of a rectangle and a semicircle. The architect decided that the height of the rectangle should be 3 feet less than the length of the diameter. The architect uses the expression $2x(2x-3)+\frac{\pi}{2}x^2$ to represent the area of the window.



not to scale

   a. What does x represent in the expression?

   b. What does the term $\frac{\pi}{2}x^2$ represent in this situation?

   c. What does the expression $2x(2x-3)$ represent?

2. The equation $y=-2(x-20)(x+1)$ models the sales, in hundreds of millions of dollars, of compact discs for years since 1990.

   a. What does the quantity $(x-20)$ indicate about the context?

   b. What does the quantity $(x+1)$ indicate about the context?

3. The population of Volusia County for the 20th century $(1900 - 2000)$ can be modeled by the equation $y = 11{,}354(1.039)^x$, where x is the number of years since 1900.

   a. What does the quantity $11{,}354$ represent?

   b. What is the constant percent rate of change?

4. Greenlings are a group of fish that share common features. The equation $y = -10(x-5)^2 + 294$ models the number of Greenlings caught, in thousands, in the United States since 2006, where x represents the number of years since 2006.

   Complete the table.

Quantity	Explanation
$x - 5$	
294	

13.9.3 Collaboration: Parts of an Equation or Expression

Work with your partner to answer the following.

1. To make a 12% acid solution, a chemist mixes 2 parts of 3% acid solution with a 16% acid solution. The expression $2(0.03x) + 0.16y$ represents the mixture, where x is the number of ounces of the 3% solution in one part and y is the number of ounces of the 16% solution in one part.

   Explain what each quantity means.

Quantity	Explanation
$(0.03x)$	
$2(0.03x)$	
$0.16y$	

2. Since 2010, the number of terawatts (1 trillion watt hours) of electricity generated from solar power can be modeled by the equation $y = 44.959(1.386)^x$.

   a. What does the variable x represent?

   b. What does the variable y represent?

   c. What does the quantity 44.959 represent?

   d. What is the constant percent rate of change?

3. The equation $y = -3(x-3)^2 + 32$ models the height, y, in feet, of a model rocket, x, seconds after taking off.

   Maya completed the explanations for the quantities in the equation.

Quantity	Explanation
$x - 3$	The number of seconds away the model rocket's height is from the maximum.
32	The height of the model rocket when it is ignited.

Discuss with your partner Maya's explanations. One of her explanations has an error. Determine the error, then write Maya a note explaining it.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

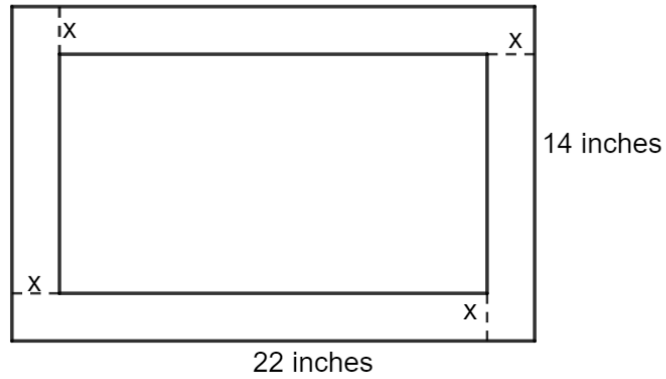
13.9.4 Try It: Parts of An Equation or Expression

1. Juanita is buying decorative rock for her garden. She is using a coupon that gives her the first 8 pounds for free. Each pound of decorative rock costs $\$0.28$ per pound. Juanita paid $\$2.40$ in sales tax.

   The expressions shown are equivalent. Circle the expression that most clearly shows the 8 pounds of rock that Juanita got for free.

   $0.28x + 0.16$ $\qquad$ $0.28x - 2.24 + 2.40$ $\qquad$ $0.28(x - 8) + 2.40$

2. A company is experimenting with the size of a gift box without a top that can be made from a 14 inch by 22 inch rectangular piece of cardboard, as shown.



   The company's mathematician uses the function $A(x) = 2x(22 - 2x) + 2x(14 - 2x) + [(22 - 2x)(14 - 2x)]$ to model the surface area in square inches, in.^{2}, of the interior of the gift box. Interpret each quantity in the table.

Quantity	Explanation
$14 - 2x$	
$22 - 2x$	
$2x(14 - 2x)$	
$(22 - 2x)(14 - 2x)$	

13.9.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson?

Unit 13, Lesson 10: Determining Which Function Type Best Models a Real-World Context

Benchmark

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.

Learning Target

☐ I can determine whether a linear, quadratic, or exponential function best models a given real-world situation.